

Customer Shopping Behavior Analysis

Data cleaning · SQL analysis · dashboarding · business recommendations

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across four product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Customer demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- **Missing data:** 37 values in the Review Rating column

3. Data Preparation in Python

- **Data loading:** imported the dataset using pandas
- **Initial exploration:** used `df.info()` and `.describe()` to check structure and summary statistics
- **Missing data handling:** imputed missing Review Rating values using the median rating of each product category
- **Column standardization:** renamed columns to snake_case for readability
- **Feature engineering:** created `age_group` (quartile-based bins) and `purchase_frequency_days`
- **Data consistency check:** verified `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`
- **Database integration:** loaded the cleaned DataFrame into PostgreSQL for SQL analysis

4. Data Analysis Using SQL

Ten business questions were answered directly against the cleaned dataset.

4.1 Revenue by Gender

Gender	Revenue (USD)
Male	\$157,890
Female	\$75,191

Male customers generate roughly double the revenue of female customers.

4.2 High-Spending Discount Users

839 customers used a discount and still spent above the overall average purchase amount of \$59.76 — showing discounts don't always signal price-sensitive buyers.

4.3 Top 5 Products by Rating

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4.4 Shipping Type Comparison

Shipping Type	Avg. Purchase Amount
Standard	\$58.46
Express	\$60.48

Express shipping customers spend about 3.5% more on average than Standard.

4.5 Subscribers vs. Non-Subscribers

Subscription	Customers	Avg. Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

Non-subscribers spend marginally more on average; subscribers contribute 27% of total revenue.

4.6 Most Discount-Dependent Products

Product	Share of Purchases Discounted
Hat	50.0%
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%
Pants	47.4%

4.7 Customer Segmentation

Segment	Customers	Share
Loyal	3,116	80%
Returning	701	18%
New	83	2%

4.8 Top 3 Products per Category

Category	Top Products (by orders)
Accessories	Jewelry (171), Sunglasses (161), Belt (161)
Clothing	Blouse (171), Pants (171), Shirt (169)
Footwear	Sandals (160), Shoes (150), Sneakers (145)
Outerwear	Jacket (163), Coat (161)

4.9 Repeat Buyers and Subscription

Among customers with more than 5 previous purchases, 958 subscribe and 2,518 do not — showing no strong link between repeat buying and subscribing.

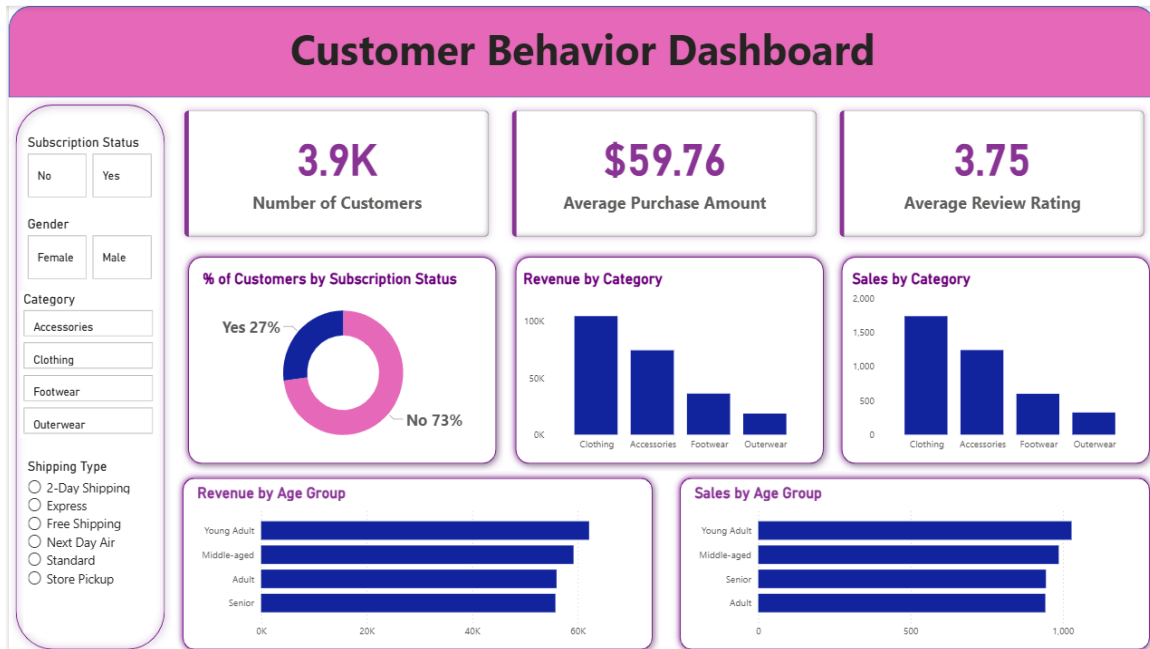
4.10 Revenue by Age Group

Age Group	Total Revenue
Young Adult	\$62,143
Middle-aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Revenue is fairly evenly spread across age groups, with Young Adult leading.

5. Dashboard in Power BI

An interactive dashboard was built in Power BI to present these insights visually, with filters for subscription status, gender, category, and shipping type.



Customer Behavior Dashboard — Power BI

6. Business Recommendations

- **Boost subscriptions** — subscribers currently spend about the same as non-subscribers, so the program isn't yet driving extra spend; exclusive perks could change that.
- **Convert New → Returning → Loyal** — only 2% of customers are "New," suggesting strong retention once someone makes a second purchase.
- **Review discount policy** — several products (Hat, Sneakers, Coat) rely on discounts for roughly half their sales; balance sales boosts with margin control.
- **Product positioning** — highlight top-rated and best-selling products in campaigns.
- **Targeted marketing** — focus efforts on the Young Adult segment and express-shipping users.